

ADDITIONAL PROJECT

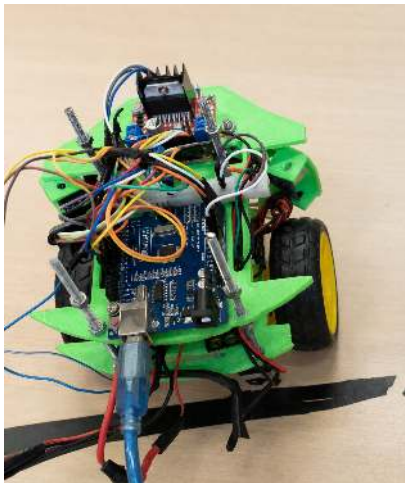
Autonomous Line-Follower Robot

Multi-Sensor Track Navigation, Obstacle Avoidance, and Color Checkpoints

CONTEXT	ROLE	PLATFORM	VALIDATION
Team Robotics Project	Embedded Development	Arduino Mega	Physical Prototype

Project Overview

Contributed to the embedded development of a team-built differential-drive robot that follows a marked track, detects forward obstacles, selects a clearer bypass direction, reacquires the line, and responds to colored checkpoints. A five-channel IR array provides the lateral error for PD steering; a servo-mounted HC-SR04 supports left/right clearance comparison; and a TCS34725 sensor triggers programmed checkpoint stops.



Physical Arduino Mega line-following robot during hardware integration and bench testing.

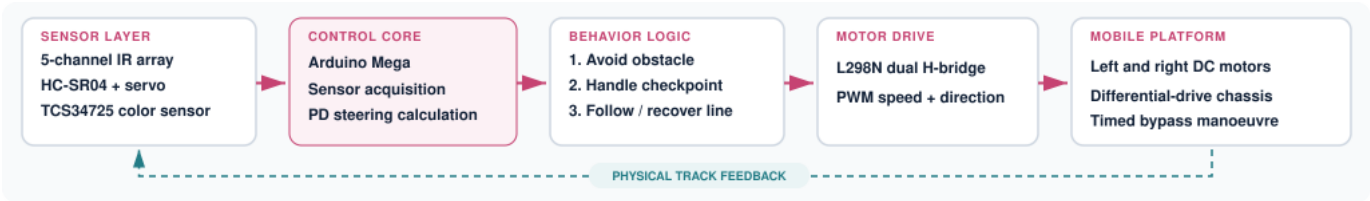
Engineering contributions

- Integrated line, distance, and color sensing with an Arduino Mega and a dual-channel L298N motor driver.
- Developed weighted five-sensor line estimation and differential motor-speed correction using a PD steering law.
- Implemented priority-based behavior arbitration: collision avoidance, checkpoint handling, then normal line following.
- Programmed servo-assisted obstacle scanning, timed detours, lost-line recovery, and automatic return to the track.

CORE RESPONSIBILITY

Embedded integration from sensor acquisition and behavior arbitration to dual-motor actuation.

SYSTEM ARCHITECTURE



Control flow: IR array, ultrasonic scan, and color sensing into Arduino decision logic, L298N drive, and differential wheel actuation.

Implementation and validation

- **Line control:** sensor weights $-4, -2, 0, 2, 4$ produce a lateral error; proportional and derivative terms correct wheel PWM.
- **Obstacle response:** a 15 cm software threshold initiates left/right scanning, timed bypass, and line reacquisition.
- **Checkpoint logic:** red and green markers command 5 s stops; blue commands a 2 s stop.

VALIDATION BOUNDARY

The physical prototype and integrated behavior are documented. No quantitative accuracy, speed, clearance, or repeatability measurements were supplied, so no numerical performance claim is made.

PROPOSED 2026 ARCHITECTURE EVOLUTION

— NOT YET IMPLEMENTED

- **Closed-loop motion:** add wheel encoders and an IMU so velocity and turn angle replace battery- and surface-dependent timing.
- **Deterministic firmware:** migrate to an STM32G4 with timer-driven acquisition, a non-blocking state machine, watchdog supervision, and explicit fault recovery.
- **Robust perception:** use a calibrated analog reflectance array and solid-state multi-zone ToF ranging to improve track estimation and remove the scanning servo.

Line control and safety-critical actuation remain deterministic on the micro-controller; AI adds no justified value to this task.

TECHNOLOGIES

Arduino Mega

Embedded C/C++

PD Control

PWM

L298N

IR Sensor Array

HC-SR04

Servo Scanning

TCS34725

I²C

Differential Drive